

Write-Up: Trader Performance vs Market Sentiment

Methodology

Two datasets were used: 211,224 Hyperliquid trades from 32 traders across 2024, and the daily Bitcoin Fear/Greed Index. After parsing timestamps, both datasets were merged at the date level **before** aggregation — so every trade row already carried its day's sentiment label prior to any groupby. This avoided a secondary join and kept the pipeline clean.

Daily per-trader metrics were then aggregated: total PnL, trade count, win rate, average position size, total volume, long/short ratio (derived from the Direction column using `str.contains` to handle all variants — "Open Long", "Close Long", etc.), and a drawdown proxy (largest single-trade loss in a day). Since no leverage column existed, a leverage proxy was computed as each trader's average position size relative to the dataset median.

Sentiment was simplified from 5 labels (Extreme Fear → Fear, Extreme Greed → Greed) to 3 (Fear, Neutral, Greed). All Fear vs Greed comparisons excluded Neutral days. Traders were segmented three ways: by position size (Low/Mid/High, tercile split), by trade frequency (Frequent/Infrequent, median split), and by performance archetype (Consistent Winner, Inconsistent, Loser based on total PnL and win rate).

For the bonus, a Random Forest classifier (200 trees, max depth 6, balanced class weights) was trained to predict whether a given trader-day would be profitable, using behavior + sentiment features. KMeans clustering ($k=3$, chosen via elbow method) on scaled trader-level features identified three behavioral archetypes, visualized via PCA projection.

Insights

1. Fear days produce more activity, not more caution. Traders averaged 105 trades/day on Fear days vs 77 on Greed days — a 37% increase that is statistically significant ($p = 0.002$). Position sizes were also 43% larger and total volume more than doubled. Contrary to the intuition that fear makes traders cautious, these traders became more aggressive. This overtrading during fear coincides with a 41% higher drawdown proxy (\$462 vs \$328), meaning the extra activity comes with meaningfully more risk.

2. Sentiment edge depends on which trader you are. High-size traders earned \$9,847/day on Fear days but only \$2,640 on Greed days — a 3.7× difference. Low-size traders showed the opposite: \$2,910 on Greed vs \$599 on Fear. Each size tier has a clear sentiment "home turf." PnL differences between Fear and Greed across all traders were not statistically significant ($p = 0.45$), meaning the effect is real but segment-specific — it washes out in the aggregate.

3. Winners buy fear; losers short it. Consistent Winners held a long ratio of 0.605 on Fear days (vs 0.478 on Greed) — they leaned bullish when the market panicked. Losers did the opposite, with a long ratio of just 0.177 on Fear days, essentially betting against recovery. This is the starkest behavioral difference between the top and bottom performer segments and aligns with the contrarian "buy the dip" principle.

Strategy Recommendations

Strategy 1 — Trade with your sentiment edge, not against it. High-size (high-leverage-proxy) traders should concentrate larger positions on Fear days, where their average PnL is 3.7× higher. Low-size traders should do the opposite — scale up only on Greed days. Practically: before sizing up a position, check the day's Fear/Greed classification and compare it to your historical segment performance. Trading on your unfavorable sentiment day at reduced size preserves capital without requiring you to sit out entirely.

Strategy 2 — Frequent traders must enforce a Fear-day trade cap. High-frequency traders see drawdowns 29% larger on Fear days (\$613 vs \$474) with no improvement in win rate (35.7% Fear vs 36.3% Greed). The extra 37% trades on Fear days add risk without adding edge. A concrete rule: on any Fear-classified day, cap daily trade count at your personal Greed-day average. Combined with Insight 3, frequent traders should also resist the urge to go short during fear — Winners go long on Fear days and outperform those who don't.